LAB EXPERIMENT
ITA0506-COMPUTER VISION

PAVENDHAN A -192321109

EXPNO: 39. Vehicle Detection Using OpenCV

Aim

To detect moving vehicles in a video using **OpenCV**.

Algorithm

1. Import OpenCV.
2. Open the car video **VEHICLE.mp4**.
3. Read each video frame.
4. Apply background subtraction.
5. Remove small noise using morphological operations.
6. Find contours of moving objects.
7. Draw rectangles around large moving objects.
8. Display the detected vehicles.
9. Press **ESC** to stop.

CODE:

EXP-39.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...

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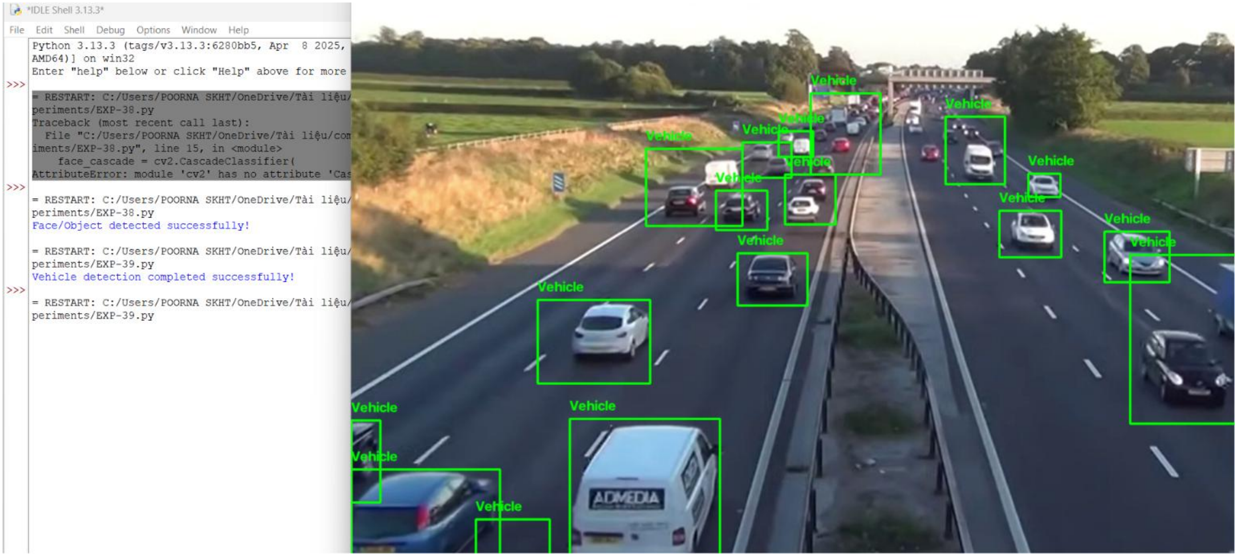
```python
    # Detect vehicles
    for contour in contours:

        area = cv2.contourArea(contour)

        if area > 500:

            x, y, w, h = cv2.boundingRect(contour)

            # Draw rectangle
            cv2.rectangle(
                frame,
                (x, y),
                (x + w, y + h),
                (0, 255, 0),
                2
            )

            # Display label
            cv2.putText(
                frame,
                "Vehicle",
                (x, y - 10),
                cv2.FONT_HERSHEY_SIMPLEX,
                0.6,
                (0, 255, 0),
                2
            )

    # Show result
    cv2.imshow("Vehicle Detection", frame)

    # Press ESC to exit
    if cv2.waitKey(30) & 0xFF == 27:
        break

cap.release()
cv2.destroyAllWindows()

print("Vehicle detection completed successfully!")
```

Ln: 70 Col: 0

OUTPUT :



Result

The moving vehicles in VEHICLE.mp4 are successfully detected and marked with rectangular bounding boxes using OpenCV.